

2016.0 RANGE ROVER (LG), 501-11

GLASS, FRAMES AND MECHANISMS

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the Glass, Frames and Mechanisms, refer to the relevant Description and Operation section in the workshop manual.

REFER to: [Glass, Frames and Mechanisms](#) (501-11 Glass, Frames and Mechanisms, Description and Operation).

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Refer to Section 100-00 General Information for window glass health and safety precautions.

1. Verify the customer concern

2. Visually inspect for obvious signs of damage and system integrity

Visual Inspection

ELECTRICAL
▪ Physical damage to the windshield

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step

4. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index

5. Check DDW for open campaigns. Refer to the corresponding bulletins

and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required

WARRANTY REPAIRS

NOTES:

- The warranty period for the windshield is twelve months with the exception of delamination and electrical faults.
- Warranty repairs should be completed using genuine parts, in accordance with the Warranty Policy and Procedures Manual.

1. Draw a line around the windshield damage using a marker pen.
2. Photograph the entire windshield. If the damage extends behind any trim, remove the trim and take further photographs.
3. Photograph the trademark logo and code to validate the windshield as factory fitment.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Scratches	▪ Debris trapped under a wiper blade ▪ Foreign object damage ▪ Fouling by trim	▪ GO to Pinpoint Test A.
Chips	▪ Foreign object damage	▪ GO to Pinpoint Test B.
Cracks	▪ Foreign object damage ▪ Impact damage during assembly	▪ GO to Pinpoint Test C.
Delamination	▪ Manufacturing defect	▪ GO to Pinpoint Test D.

PINPOINT TESTS

NOTE:

A scratch will usually be regular in shape, following the line of the object that caused it.

PINPOINT TEST A : SCRATCH TESTS

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
A1: SCRATCH TEST 1	
	1 Probe using the tip of a pencil to identify a groove in the windshield surface.
	Is there a groove? Yes Windshield scratched. GO to A2. No Defect not valid.
A2: SCRATCH TEST 2	
	1 Check for trim, body panels, or foreign objects that may have caused the scratch.
	Was the scratch caused by a foreign object? Yes The damage is not due to a defect or an assembly error. No Rectify as appropriate.

NOTE:

Impact damage may cause a crack to form.

PINPOINT TEST B : CHIP TESTS

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
B1: CHIP TEST 1	
	1 Assess the damage by probing with the tip of a pencil.
	Is the damaged area rough (indicating a breach of the windshield surface)? Yes Damage caused by the impact of a foreign object. Not a manufacturing defect. No Install a new windshield.

NOTE:

Multiple cracks will radiate out from the source.

PINPOINT TEST C : CRACK TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
C1: CRACK TEST 1	
	NOTE: A crack will be detectable as a step in the glass. 1 Confirm the presence of a crack by probing with the tip of a pencil.
	Is the windshield cracked? Yes Windshield cracked. GO to C2. No Windshield not cracked. GO to Pinpoint Test A.
C2: CRACK TEST 2	
	1 Assess the source of the crack by probing with the tip of a pencil.
	Is there evidence of impact damage being the source of the crack? Yes GO to Pinpoint Test B. No

Install a new windshield.

PINPOINT TEST D : DELAMINATION TESTS

TEST CONDITIONS

DETAILS/RESULTS/ACTIONS

D1: DELAMINATION TEST 1



1 Visually assess the windshield for delamination.

Have the glass laminates separated?

Yes

Install a new windshield.

No

No further action.

DTC INDEX

For a complete list of all Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.